

Final Research Report (internal)

Workspace: /Volumes/PRO-BLADE/Wharton-Experiments **Revised:** 2026-09-27 (integrity audit; supersedes the 2026-09-26 version) **Status:** Legacy lean matrix complete (1181/1181) but **leakage-contaminated**. Lockbox opened once; H1 not supported on valid evidence. Repaired exploratory pipeline: **no forecasting skill over the historical mean**. Client IPS still BLOCKED.

1. Summary

The 2026-09-26 report selected a ridge forecast after it "clearly beat persistence" on a virgin synthetic lockbox. That result is an artefact: the synthetic features contain the contemporaneous driver of the label (D-050). After repairing the data path (causal features, genuinely disjoint folds), neither the frozen ridge nor a standardised ridge beats the historical mean on real ETF data or on any synthetic seed. Beating random-walk persistence is uninformative, because any forecast shrunk toward zero does so. The defensible decision is to **not rely on return forecasts**: use a strategic allocation with EWMA risk sizing and an operating-cash reserve.

Distinctions: implemented $\neq$ validated $\neq$ investable; lockbox $\neq$ funding guarantee; proposed trade $\neq$ WInS execution.

2. Evidence generations

Generation	Label	Use
Pre-lean receipts (1916 orphans under runs/)	historical / superseded	Excluded from all tables (D-055)
Lean matrix, 1181 cells (protocol/EXPERIMENT_MANIFEST.csv)	leakage-contaminated	Engineering smoke evidence only
Lockbox, single open (runs/lockbox/)	preregistered, frozen	Synthetic arm invalid; ETF arm negative
Factorial (runs/factorial/)	invalid (look-ahead, forecast unused)	Provenance only (D-054)
Repaired pipeline (runs/repaired/)	exploratory, post-lockbox	Current best evidence; not confirmatory

3. Defects found in the legacy pipeline

- Synthetic leak (D-050).** Track C features $[\text{latent}_t, \text{latent}_{\{t-1\}}]$, label $\beta \cdot \text{latent}_t + \epsilon, \epsilon \sim N(0, 0.01^2)$. Ridge recovers the label up to noise: lockbox MSE $9.89\text{e-}5 \approx 1\text{e-}4 = \text{Var}(\epsilon)$.
- ETF leak (D-051).** Expanding-std feature at row t includes r_t . Every tabular "h20" label is actually a 1-step return.
- Degenerate replication (D-052).** All legacy folds share one split at lean sizes, and ETF seeds share one panel. In the full profile each model has 1 distinct ETF value and ≤ 5 distinct synthetic values across 20 cells. "Validation folds 0–2" are the same split as full fold 3.
- Mislabelled target (D-053).** ETF asset 0 is AGG (bond ETF), not SPY.
- Mixed generations (D-055).** Scoreboards previously averaged 1916 orphan receipts into the full profile ($n=120$ per model vs 40 in the manifest).

4. Legacy results with corrected status

Selection rule (preregistered): min pooled MSE, prefer B_RIDGE within 1%. M04 $5.356\text{e-}05$, B_RIDGE $5.365\text{e-}05$. Selection stands as a recorded decision, but it is **not evidence of skill** (pooled MSE is dominated by the leaky synthetic track, and the folds are not distinct).

Lockbox dataset	B_RIDGE MSE	Persistence MSE	Status
synthetic seed 997	9.89e-05	0.0281	INVALID — leakage
ETF offset 800 (AGG)	2.14e-05	2.12e-05	NEGATIVE — ridge not better

- H1: not supported on valid evidence. H2 (finite EWMA Frobenius error): true but trivial.
- Multiplicity over the legacy full profile (Wilcoxon vs persistence, BH): after collapsing duplicate cells, **0/10 comparisons are testable** (ETF 1 unique pair, synthetic 5).
reports/evidence_bank/MULTIPLICITY_SUMMARY.md.
- M11 funding_shortfall is 0.0 in every cell (the obligation never binds), so it is uninformative.

5. Repaired exploratory evaluation (D-056)

Setup: label = next-day return of asset 0. Causal features: lag-1 return, 5-day mean, 20-day std, and lag-1 cross-sectional mean, all computed from rows < t. Folds: expanding window, 8 × 100 disjoint test rows, minimum 300 training rows. ETF: 1691-row cache (sha256 6c234590...); the one fold overlapping lockbox rows [800, 960) is dropped, leaving 700 test days. Synthetic: SIGNAL world, frozen full seeds {11, 23, 47, 89, 131}, 1200 steps, 800 test days each. Statistics: Diebold–Mariano (Newey–West) on squared-error differentials, BH over the 48-test challenger family, and a fold-level Wilcoxon. Git HEAD at run: 2b14666.

Pre-declared primary contrast, standardised ridge vs historical mean ($\text{OOS } R^2 = 1 - \text{MSE_ridge} / \text{MSE_mean}$):

Dataset	OOS R^2	DM p	BH q
ETF (AGG)	+0.22%	0.742	0.810
synthetic s11	−0.62%	0.221	0.342
synthetic s23	+0.14%	0.848	0.884
synthetic s47	−0.51%	0.117	0.224
synthetic s89	−0.23%	0.621	0.753
synthetic s131	+0.19%	0.681	0.765

- No ridge-vs-mean or ridge-vs-zero contrast is significant after BH (0/24).
- Ridge vs 1-step persistence: OOS R^2 +46% to +50% on every dataset ($q < 1e-20$). This reflects the weakness of persistence (for near-i.i.d. returns its $\text{MSE} \approx 2 \times \text{variance}$), not skill.
- Leakage demonstration: the frozen B_RIDGE on legacy features over the same test rows gets OOS $R^2 \approx +99.5\%$ on every seed.
- The frozen B_RIDGE (unstandardised, $\alpha = 1$, no intercept) is numerically close to a zero forecast on return-scale features.

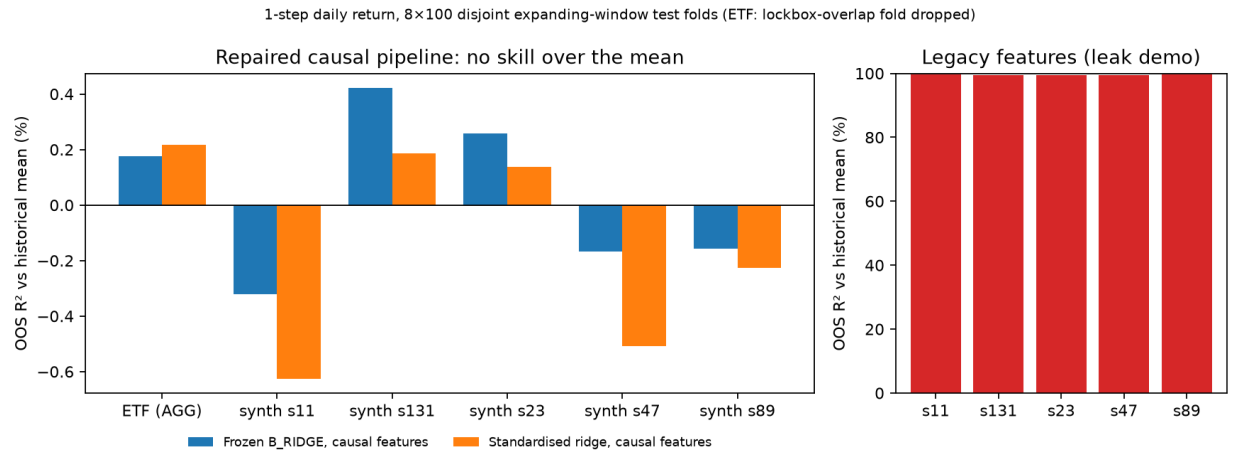


Figure: scripts/figures/repaiored_oos_r2.py; source data reports/figures/repaiored_oos_r2.csv. Full tables: reports/evidence_bank/REPAIRED_EVAL_SUMMARY.md.

6. Limitations

- The repaired evaluation is post-lockbox and exploratory. ETF training windows include rows that were in the spent lockbox window (test windows exclude them).
- Only tabular baselines were re-evaluated. M01–M12 were **NOT RUN** on the repaired pipeline, so the legacy ranking of complex models is unsupported either way.
- The synthetic SIGNAL world has an i.i.d. latent, so no causal predictability exists by construction. It validates the pipeline (no spurious skill), not a forecasting method.
- The ETF cache is not versioned in git. A fresh download will differ (see data/manifests/DATA_MANIFEST.yaml).
- The target is a single asset (AGG) at a 1-step horizon. There are no transaction costs in the forecasting evaluation.

7. Blocked

WGY case PDFs, WInS executions, client wealth/obligations, Track B point-in-time data, submission authority.

8. Reproduce

```
bash scripts/bootstrap.sh
PYTHONPATH=src .venv/bin/python scripts/run_repaired_eval.py
PYTHONPATH=src .venv/bin/python scripts/run_multiplicity.py
.venv/bin/python scripts/figures/repaiored_oos_r2.py
bash scripts/build_reports.sh && .venv/bin/python scripts/build_evidence_bank.py
.venv/bin/python scripts/build_pdfs.py
```